

Commutative Algebra & Algebraic Geometry

Exercise 9

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9.1 Problem

- (1) Prove that the homogenization and dehomogenization of polynomials are inverses, except for losing the highest x_0 factor.
- (2) Consider $I = (f, g) \leq k[x, y, z]$ where $f = y - x^2, g = z - x^3$. Show that the homogenization $I_h \leq k[x, y, z, w]$ does not equal (f_h, g_h) by showing that $xy - zw \in I_h - (f_h, g_h)$.
- (3) Let $f \in k[x_0, \dots, x_n]$ be an irreducible homogeneous polynomial. Show that the degree of $Z_{\mathbb{P}^n}(f)$ is independent of f .

- (1) For a multi-index $\alpha = (\alpha_1, \dots, \alpha_n)$ and indeterminants $x = (x_1, \dots, x_n)$ define

$$x^\alpha = \prod_{i=1}^n x_i^{\alpha_i}.$$

Let $x/x_0 = (x_1/x_0, \dots, x_n/x_0)$, then for $f \in k[x_1/x_0, \dots, x_n/x_0]$ we can write it as

$$f = \sum_{\alpha} a_{\alpha} (x/x_0)^{\alpha},$$

where the sum ranges over a finite set of multi-indices and $a_{\alpha} \in k$. We can then order the sum by degree of multi-index (where $|\alpha| = \sum \alpha_i$)

$$f = \sum_{n=0}^d \sum_{|\alpha|=n} a_{\alpha} (x/x_0)^{\alpha} = \sum_{n=0}^d \sum_{|\alpha|=n} a_{\alpha} \frac{x^{\alpha}}{x_0^n}.$$

Thus f 's homogenization is

$$f_h = \sum_{n=0}^d \sum_{|\alpha|=n} a_{\alpha} x_0^{d-n} x^{\alpha}.$$

The dehomogenization of this is

$$(f_h)_{dh} = f_h(1, x/x_0) = \sum_{n=0}^d \sum_{|\alpha|=n} a_{\alpha} (x/x_0)^{\alpha} = f,$$

as desired.

Now for $g \in k[x_0, \dots, x_n]$ homogeneous of degree d we write

$$g = \sum_{\beta} b_{\beta} \hat{x}^{\beta}$$

for $b_{\beta} \in k$, $\hat{x} = (x_0, \dots, x_n)$, and all the degrees of the multi-indices in the sum are equal to d . For multi-index β , we can write it as $\beta = (n, \alpha)$ for multi-index α over $(x_1, \dots, x_n)$. Since $|\beta| = n + |\alpha| = d$, we have that $n = d - |\alpha| \leq d$. So we can then rewrite g as (writing $b_{\alpha} = b_{(d-n, \alpha)}$):

$$g = \sum_{n=0}^d \sum_{|\alpha|=n} b_{\alpha} x_0^{d-n} x^{\alpha}.$$

Its dehomogenization is

$$g_{\text{dh}} = g(1, x/x_0) = \sum_{n=0}^d \sum_{|\alpha|=n} b_{\alpha} \frac{x^{\alpha}}{x_0^n}.$$

This is a polynomial whose degree is the maximal N such that there is a multi-index where $|\alpha| = N$ and $b_{\alpha} \neq 0$. Then

$$(g_{\text{dh}})_{\text{h}} = \sum_{n=0}^d \sum_{\alpha} b_{\alpha} x_0^{N-n} x^{\alpha}.$$

So we see that

$$g = x_0^{d-N} (g_{\text{dh}})_{\text{h}},$$

and x_0 does not divide $(g_{\text{dh}})_{\text{h}}$, since for α of degree N with $b_{\alpha} \neq 0$, in $(g_{\text{dh}})_{\text{h}}$ its component is $b_{\alpha} x^{\alpha}$ which is not divided by x_0 ; so we have removed the highest factor of x_0 from g .

(2) Firstly, we see

$$(xf - g)_{\text{h}} = (xy - x^3 - z + x^3)_{\text{h}} = (xy - z)_{\text{h}} = xy - zw,$$

and since $xf - g \in I$ we see that $xy - zw \in I_{\text{h}}$.

We see that

$$f_{\text{h}} = wy - x^2, \quad g_{\text{h}} = w^2 z - x^3.$$

And $xy - zw$ cannot be in the ideal generated by these. Indeed, multiplying any monomial by f_{h} or g_{h} cannot yield an xy term: $X^{\alpha}wy$ and $X^{\alpha}x^2$ both are not xy for example. Thus multiplying any polynomial by f_{h} or g_{h} yields a sum of monomials, none of which are xy , and so the sum itself cannot yield an xy term. And so the sum of products of polynomials with $f_{\text{h}}, g_{\text{h}}$ cannot yield an xy term, and thus cannot be equal to $xy - zw$.

(3) Let $f, g \in k[x_0, \dots, x_n]$ be irreducible homogeneous polynomials with $Z_{\mathbb{P}^n}(f) = Z_{\mathbb{P}^n}(g)$. We want to show that they have the same degree. In question 2 we will show that if $Z_{\mathbb{P}^n}(f)$ is nonempty then $\mathcal{I}(Z_{\mathbb{P}^n}(f)) = \sqrt{(f)}$. Since f is irreducible, this is equal to (f) . So in the case that the zerosets are nonempty we have $(f) = (g)$ so $f = gp$ and $g = fq$ for polynomials p, q . But then $\deg(f) = \deg(g) + \deg(q)$ and $\deg(g) = \deg(f) + \deg(q)$, meaning $\deg(f) \leq \deg(g) \leq \deg(f)$ so we have equality.

9.2 Problem

Let $n \in \mathbb{N}$.

(1) The **irrelevant ideal** of $k[x_0, \dots, x_n]$ is

$$k[x_0, \dots, x_n]_{+} = \bigoplus_{d>0} k[x_0, \dots, x_n]_d.$$

Let $I \leq k[x_0, \dots, x_n]$ be a homogeneous ideal. Show that the following are equivalent:

$$(1) Z_{\mathbb{P}^n}(I) = \emptyset, \quad (2) k[x_0, \dots, x_n]_{+} \leq \sqrt{I}, \quad (3) k[x_0, \dots, x_n]_d \subseteq I \text{ for some } d.$$

Such an ideal is called an **irrelevant ideal** (as opposed to *the* irrelevant ideal).

(2) Let $J \leq k[x_0, \dots, x_n]$ be a homogeneous ideal which is not irrelevant. Show that $\mathcal{I}(Z_{\mathbb{P}^n}(J)) = \sqrt{J}$.
Conversely show that for $Y \subseteq \mathbb{P}^n$ closed, $Z_{\mathbb{P}^n}(\mathcal{I}(Y)) = \overline{Y}$.

(3) Let $\text{cl}(\mathbb{P}^n)$ be the closed subsets of $\mathbb{P}^n$ and rad_{h} be the radical homogeneous non-irrelevant ideals of $k[x_0, \dots, x_n]$. We have maps

$$\mathcal{I}: \text{cl}(\mathbb{P}^n) \rightleftarrows \text{rad}_{\text{h}}: Z.$$

Show the following.

- (i) $\mathcal{I}, Z$ form an antiequivalence of posets.
 - (ii) They send irreducible closed subsets to prime ideals and vice versa.
 - (iii) The maps interchange points $[a_0 : \cdots : a_n]$ and ideals $\langle a_i x_j - a_j x_i \rangle_{i,j}$.
- (4) Show that $\mathbb{P}^n$ is irreducible.
- (5) Show that $\mathbb{P}^n$ is a Noetherian topological space of dimension n .

- (1) Since the projection $\pi: \mathbb{A}^{n+1} - 0 \rightarrow \mathbb{P}^n$ is surjective, $Z_{\mathbb{P}^n}(I)$ is empty iff its preimage is. Thus

$$Z_{\mathbb{P}^n}(I) = \emptyset \iff \emptyset = \pi^{-1} Z_{\mathbb{P}^n}(I) = Z_{\mathbb{A}^{n+1}-0}(I) \iff Z_{\mathbb{A}^{n+1}}(I) \subseteq \{0\}.$$

We know that $\{0\} = Z_{\mathbb{A}^{n+1}}(x_0, \dots, x_n)$, so this is iff

$$\sqrt{(x_0, \dots, x_n)} \leq \sqrt{I},$$

and $(x_0, \dots, x_n)$ is maximal so it is radical. Furthermore, $k[x_0, \dots, x_n]_+ = (x_0, \dots, x_n)$ because the irrelevant ideal clearly contains this maximal ideal as x_i are homogeneous of degree 1, and the irrelevant ideal is proper so there is equality by maximality.

So we have shown $Z_{\mathbb{P}^n}(I) = \emptyset$ iff $k[x_0, \dots, x_n]_+ \leq \sqrt{I}$, which is the equivalence of (1) and (2).

If $k[x_0, \dots, x_n]_d \leq I$ for any d then $Z_{\mathbb{P}^n}(I)$ is empty. This is because it implies $x_0^d, \dots, x_n^d \in I$ and their only common zero is 0. So $Z_{\mathbb{A}^{n+1}}(I) = \{0\}$ and thus $Z_{\mathbb{P}^n}(I) = \emptyset$. So (3) implies (1).

Conversely suppose (1), so we get (2) by equivalence. So $x_0, \dots, x_n \in \sqrt{I}$ and thus there are $k_0, \dots, k_n$ such that $x_0^{k_0}, \dots, x_n^{k_n} \in I$. Let $d = k_0 + \cdots + k_n$, then $k[x_0, \dots, x_n]_d \subseteq I$. Indeed if α is any multi-index of degree d , then there is some $\alpha_i \geq k_i$ so $x_i^{k_i} \in I$ occurs in the product for x^α , so $x^\alpha \in I$. So then any linear combination of monomials of degree d , i.e. an element of $k[x_0, \dots, x_n]_d$, is in I as desired.

- (2) Since J is homogeneous, $J = \bigoplus_d k[\bar{x}]_d \cap J$, and since it is not irrelevant it cannot contain any scalars (as then it would have empty zero set), so this sum can be taken over $d > 0$ showing that J is contained in the irrelevant ideal. Thus $J \leq (x_0, \dots, x_n)$.

Now we claim

$$\mathcal{I}(Z_{\mathbb{P}^n}(J)) = \mathcal{I}_{\mathbb{A}}(Z_{\mathbb{A}^{n+1}}(J)) = \sqrt{J}.$$

The left-hand side is generated by homogeneous f such that for every $P \in Z_{\mathbb{P}^n}(J)$, $f(P) = 0$. Since $Z_{\mathbb{P}^n}(J)$ is nonempty these f are not scalars (the condition cannot be vacuous). This means that for every lift of P to $\tilde{P}$, $f(\tilde{P}) = 0$. For $\tilde{P} \in Z_{\mathbb{A}^{n+1}}(J)$, if $\tilde{P} \neq 0$ then $\pi(\tilde{P}) \in Z_{\mathbb{P}^n}(J)$ as we showed in class. Thus $f(\tilde{P}) = 0$ so $f \in \mathcal{I}_{\mathbb{A}}(Z_{\mathbb{A}}(J))$. And for $\tilde{P} = 0$, since f is homogeneous and not a scalar, $f(0) = 0$. Thus $f(\tilde{P}) = 0$ in all cases, so $f \in \mathcal{I}_{\mathbb{A}}(Z_{\mathbb{A}}(J))$, showing left-to-right inclusion.

Now $Z_{\mathbb{A}^{n+1}}(J)$ is invariant under $k^\times$'s action, since J is generated by homogeneous polynomials and their roots are invariant under $k^\times$. As we showed in class, this means $\mathcal{I}_{\mathbb{A}}(Z_{\mathbb{A}}(J))$ is homogeneous and thus generated by its homogeneous elements. For any homogeneous f in this ideal, it is also in $\mathcal{I}(Z_{\mathbb{P}}(J))$: take any $P \in Z_{\mathbb{P}}(J)$ and lift it to $\tilde{P}$. Then $\tilde{P} \in \pi^{-1} Z_{\mathbb{P}}(J) = Z_{\mathbb{A}}(J) - 0$ and so $f(\tilde{P}) = 0$ so $f(P) = 0$. So all of $\mathcal{I}_{\mathbb{A}}(Z_{\mathbb{A}}(J))$'s homogeneous elements are in $\mathcal{I}(Z_{\mathbb{P}}(J))$ and as it is generated by those, there is inclusion and thus equality.

For the converse, closed sets of $\mathbb{P}^n$ are of the form $Z_{\mathbb{P}}^n(J)$ for homogeneous ideals J . Let $Y \subseteq \mathbb{P}^n$ be nonempty, and suppose $Y \subseteq Z_{\mathbb{P}}(J)$; since Y is nonempty J is not irrelevant. Then since $\mathcal{I}$ is order-reversing, we see that from what we just proved

$$\mathcal{I}(Y) \supseteq \mathcal{I}(Z_{\mathbb{P}}(J)) = \sqrt{J} \supseteq J.$$

And so

$$Z_{\mathbb{P}}(\mathcal{I}(Y)) \subseteq Z_{\mathbb{P}}(J).$$

And $Y \subseteq Z_{\mathbb{P}}(\mathcal{I}(Y))$ clearly, so it is the smallest closed set containing Y ; its closure.

(3)

- (i) This is clear, as by the previous points Z maps non-irrelevant ideals to non-empty closed subsets, and thus

$$\mathcal{I}(Z(J)) = \sqrt{J} = J$$

for non-irrelevant radical J . And $Z(\mathcal{I}(Y)) = \overline{Y}$ so for Y closed this is just Y .

- (ii) For $\mathfrak{p}$ prime, suppose $Z(\mathfrak{p})$ is the union of two closed subsets, $Z(\mathfrak{p}) = Z(I) \cup Z(J) = Z(IJ)$. Then

$$\mathfrak{p} = \mathcal{I}(Z(\mathfrak{p})) = \mathcal{I}(Z(IJ)) = \sqrt{IJ},$$

where the final equality is because $Z(IJ)$ is nonempty (as it is equal to $Z(\mathfrak{p})$). In particular we have $IJ \leq \mathfrak{p}$, and since $\mathfrak{p}$ is prime one of them is contained in $\mathfrak{p}$. Say $I \leq \mathfrak{p}$ so $Z(I) \supseteq Z(\mathfrak{p})$ and thus we have equality. So $Z(\mathfrak{p})$ is irreducible.

Now suppose $Z(J)$ is irreducible for J radical; we claim that J is prime. Suppose $I_0 I_1 \leq J$ for ideals I_0, I_1 , then $Z(J) \subseteq Z(I_0 I_1) = Z(I_0) \cup Z(I_1)$. Since $Z(J)$ is irreducible this means that $Z(J) \subseteq Z(I_0)$ or $Z(J) \subseteq Z(I_1)$, say $Z(I_0)$. This implies, as $Z(I_0)$ is then non-empty,

$$I_0 \subseteq \sqrt{I_0} = \mathcal{I}(Z(I_0)) \leq \mathcal{I}(Z(J)) = \sqrt{J} = J.$$

Thus J is prime, as desired. Since $\mathcal{I}(Z(J)) = J$, this shows that $\mathcal{I}$ maps irreducible closed sets to prime ideals.

- (iii) Choose i such that $a_i \neq 0$. Then x is a zero for $a_i x_j - a_j x_i$ for all j means that $x_j = (a_j/a_i)x_i$. Thus

$$[x_0 : \cdots : x_n] = \left[\frac{a_0}{a_i} : \cdots : \frac{a_n}{a_i} \right] = [a_0 : \cdots : a_n],$$

so that

$$Z \langle a_i x_j - a_j x_i \rangle = \{[a_0 : \cdots : a_n]\},$$

since clearly $(a_0, \dots, a_n)$ is a root of each $a_i x_j - a_j x_i$. We claim that $I = \langle a_i x_j - a_j x_i \rangle$ is prime; this is sufficient because then as $\mathcal{I}, Z$ are inverses on non-irrelevant radical ideals this means $\mathcal{I}[a_0 : \cdots : a_n] = I$.

Take $a_i \neq 0$ and define

$$\varphi: k[x_0, \dots, x_n] \rightarrow k[t], \quad x_j \mapsto \frac{a_j}{a_i} x_i.$$

This is surjective as x_i is mapped to t . Clearly I is contained in the kernel of φ , we claim there is equality. Let $f \in \ker \varphi$, then modulo I all x_j are scalar multiples of x_i , so f is equivalent to $g(x_i)$ modulo I for some single-variable g . Since I is contained in the kernel, this means that $\varphi(f) = \varphi(g(x_i)) = 0$, but $\varphi(g(x_i)) = g(t)$ and this is zero iff $g = 0$. So f is equal to zero modulo I , meaning $f \in I$.

Thus $\ker \varphi = I$ and so φ induces an isomorphism $k[x_0, \dots, x_n]/I \cong k[t]$ which is a domain so I is prime, as desired.

- (4) $\mathbb{P}^n$ is closed and $\mathcal{I}(\mathbb{P}^n) = 0$ since the only polynomial which zeroes all of $\mathbb{P}^n$ is the zero polynomial. The zero ideal is prime, and so by the previous point this means $\mathbb{P}^n$ is irreducible.
- (5) $\mathbb{P}^n$ is an algebraic variety and those are always Noetherian. This is because it has a finite open affine cover, and affine varieties are Noetherian. A finite union of open Noetherian subsets is Noetherian: let $X = \bigcup_i U_i$ for U_i Noetherian, then for every $Y \subseteq X$ we have $Y = \bigcup_i U_i \cap Y$. Since U_i is Noetherian,

$U_i \cap Y$ is quasicompact, and thus Y is quasicompact as a finite union of quasicompact sets. So all of X 's subsets are quasicompact, meaning X is Noetherian.

$\mathbb{P}^n$ has an open affine cover by affine varieties $U_i \cong \mathbb{A}^n$. We showed that for an open cover of a topological space $X = \bigcup_i U_i$, $\dim X = \sup_i \dim U_i$. In this case

$$\dim \mathbb{P}^n = \sup_i \dim U_i = \sup_i \dim \mathbb{A}^n = \sup_i n = n,$$

as desired.

9.3 Problem

Let X be an affine variety and fix $n \in \mathbb{N}$.

- (1) Let $f_0, \dots, f_n \in k[X]$. Define $X' = X - Z(f_0, \dots, f_n)$, and define $f: X' \rightarrow \mathbb{P}^n$ by $f(P) = [f_0(P) : \dots : f_n(P)]$. Prove that this is a well-defined morphism of varieties.
- (2) Let $m \in \mathbb{N}$, $X = \mathbb{A}^{m+1}$, and let $f_0, \dots, f_n \in k[X]$ be homogeneous of the same degree such that $Z(f_0, \dots, f_n) = \{0\}$. Thus by point (1) we get a morphism $f: X - \{0\} \rightarrow \mathbb{P}^n$. Show that f induces a well-defined morphism $\bar{f}: \mathbb{P}^m \rightarrow \mathbb{P}^n$.

- (1) Note that f decomposes as

$$f = X' \xrightarrow{f'} \mathbb{A}^{n+1} - 0 \xrightarrow{\pi} \mathbb{P}^n$$

where $f'(P) = (f_0(P), \dots, f_n(P))$ and π is the projection morphism. f' is a morphism of varieties since each of its components are regular. Explicitly f' is a morphism into $\mathbb{A}^{n+1}$ because of this, and it maps into $\mathbb{A}^{n+1} - 0$ so by the universal property of the induced structure on $\mathbb{A}^{n+1} - 0$ it is a morphism to $\mathbb{A}^{n+1} - 0$. Thus f is a composition of morphisms and thus a morphism itself.

- (2) Consider the following commutative diagram

$$\begin{array}{ccc} \mathbb{A}^{m+1} - 0 & \xrightarrow{f'} & \mathbb{A}^{n+1} - 0 \\ \pi \downarrow & & \downarrow \pi \\ \mathbb{P}^m & \xrightarrow{\bar{f}} & \mathbb{P}^n \end{array}$$

Where f' is the map from the previous point and $\bar{f}$ is defined as

$$\bar{f}(P) = [f_0(P) : \dots : f_n(P)].$$

This is well-defined because for any two lifts of P , $\tilde{P}_1 = \lambda \tilde{P}_0$ we have that

$$[f_0(\tilde{P}_1) : \dots : f_n(\tilde{P}_1)] = [\lambda^d f_0(\tilde{P}_0) : \dots : \lambda^d f_n(\tilde{P}_0)] = [f_0(\tilde{P}_0) : \dots : f_n(\tilde{P}_0)],$$

where d is the common degree of the f_i s. The diagram clearly commutes: either path gives $P \mapsto [f_0(P) : \dots : f_n(P)]$ for $P \in \mathbb{A}^{m+1} - 0$.

Thus $\bar{f} \circ \pi = \pi \circ f' = f$, and since π is the quotient map, by its universal property it means that $\bar{f}$ is a morphism of varieties.

9.4 Problem

Let $f = yz^2 - x^3 + 3xz^2 - z^3$ and $g = yz - x^2 + 2xz$ in $\mathbb{C}[x, y, z]$. Denote $C = Z_{\mathbb{P}}(f)$, $D = Z_{\mathbb{P}}(g)$. Prove Bezout's theorem by hand in this case.

We first find $C \cap D = Z_{\mathbb{P}}(f, g)$. Note

$$(f, g) = (f - gz, g) = (-(x - z)^2(x + z), g) = ((x - z)^2(x + z), g).$$

So any (affine) zero has to have $x = \pm z$. If $x = z$ then we have

$$0 = g(z, y, z) = yz + z^2 \implies z = 0 \text{ or } y = -z.$$

If $x = -z$ then

$$0 = g(-z, y, z) = yz - 3z^2 \implies z = 0 \text{ or } y = 3z.$$

So the affine zeroes are

$$Z_{\mathbb{A}}(f, g) = \{(0, y, 0), (z, -z, z), (-z, 3z, z)\},$$

and thus projecting onto $\mathbb{P}^2$ gives

$$Z_{\mathbb{P}}(f, g) = \{[0 : 1 : 0], [1 : -1 : 1], [-1 : 3 : 1]\}.$$

So we need to find the multiplicity at each point.

For $P = [0 : 1 : 0]$, we take the local affine neighborhood U_1 , in which P corresponds to the point $(0, 0)$, and f, g correspond to their dehomogenizations:

$$f_{\text{dh}} = z^2 - x^3 + 3xz^2 - z^3, \quad g_{\text{dh}} = z - x^2 + 2xz.$$

And the stalk at P is $k[x, z]_{(x, z)}$. When quotienting by g_{dh} , we see that

$$z = \frac{x^2}{1 + 2x},$$

and thus

$$f_{\text{dh}} = x^3 \left(\frac{x}{(2x + 1)^2} - 1 + \frac{3x^2}{(2x + 1)^2} - \frac{x^3}{(2x + 1)^3} \right).$$

The term in the parentheses evaluates to -1 at $x = 0$, which means it is a unit as this is nonzero at P . So

$$\frac{\mathcal{O}_{\mathbb{P}^2, P}}{(f, g)} = \frac{k[x]_{(x)}}{(x^3)}.$$

This has a basis $1, x, x^2$ so the multiplicity at $[0 : 1 : 0]$ is 3.

For $P = [1 : -1 : 1]$ we take the local affine neighborhood U_2 , in which P corresponds to $(1, -1)$, and f, g correspond to their dehomogenizations:

$$f_{\text{dh}} = y - x^3 + 3x - 1, \quad g_{\text{dh}} = y - x^2 + 2x.$$

Now we can write these in terms of $x = u + 1$ and $y = v - 1$, so that P then corresponds to 0 and

$$f_{\text{dh}} = v - u^3 - 3u^2, \quad g_{\text{dh}} = v - u^2.$$

Then

$$(f_{\text{dh}}, g_{\text{dh}}) = (u^3 + 2u^2, v - u^2),$$

and $u^3 + 2u^2 = u^2(u + 2)$ where $u + 2$ is a unit because at P it is nonzero, so

$$(f_{\text{dh}}, g_{\text{dh}}) = (u^2, v - u^2) = (u^2, v).$$

Thus we have

$$\frac{\mathcal{O}_{\mathbb{P}^2, P}}{(f, g)} = \frac{k[u, v]_{(u, v)}}{(u^2, v)},$$

which has basis $1, u$; the multiplicity at $[1 : -1 : 1]$ is then 2.

For $P = [-1 : 3 : 1]$ we take local affine neighborhood U_2 , in which P corresponds to $(-1, 3)$ and f, g correspond to the same dehomogenizations as previously. Again we shift our coordinates, $x = u - 1$ and $y = v + 3$ so

$$f_{\text{dh}} = v - u^3 + 3u^2, \quad g_{\text{dh}} = v - u^2 + 4u.$$

And so

$$(f_{\text{dh}}, g_{\text{dh}}) = (v - u^2 + 4u, u^3 - 4u^2 + 4u) = (v - u^2 + 4u, u(u^2 - 4u + 4)) = (v - u^2 + 4u, u) = (v, u),$$

where the second-to-last equality is because $u^2 - 4u + 4$ is a unit in the stalk. Thus $(f_{\text{dh}}, g_{\text{dh}})$ is maximal, they intersect transversally at P and so the multiplicity is 1.

So the total sum of multiplicities is $3 + 2 + 1 = 6$. And $\deg f = 3, \deg g = 2$ so we have demonstrated Bezout's theorem.

9.5 Problem

Fix $n > 0$.

- (1) Let $h \in k[x_0, \dots, x_n]$ be a linear form, i.e. a homogeneous polynomial of degree 1. Show that the hyperplane $H = Z_{\mathbb{P}}(h)$ is isomorphic to $\mathbb{P}^{n-1}$.
- (2) Fix $P = [a_0 : \dots : a_n] \in \mathbb{P}^n$ not in H . Define $\varphi: \mathbb{P}^n - P \rightarrow H$ as follows. For $Q \in \mathbb{P}^n - P$, connect it to P by a line, then let $\varphi(Q)$ be the intersection of that line with H . Show that

$$\varphi([b_0 : \dots : b_n]) = \left[b_i - \frac{h(b_0, \dots, b_n)}{h(a_0, \dots, a_n)} a_i \right]_i.$$

I unfortunately do not have time to draw, but if I did I would draw the empty box of hot chocolate packets lying on my desk...

- (1) Let us write

$$h = \sum_i \beta_i x_i, \quad \beta_i \in k.$$

There is some i for which $\beta_i \neq 0$, potentially by changing variables we can assume $i = 0$. Let us write the coordinates of $\mathbb{P}^{n-1}$ by $[a_1 : \dots : a_n]$. Now consider $\Phi: \mathbb{A}^n - 0 \rightarrow \mathbb{P}^n$ given by question 3 and the polynomials $-\frac{1}{\beta_0} \sum_{i>0} \beta_i x_i$ and $x_1, \dots, x_n$; we write it as $\Phi = \pi \circ \Phi'$ as in question 3. These are homogeneous of the same degree so by question 3 part 2, this induces a morphism $\bar{\Phi}: \mathbb{P}^{n-1} \rightarrow \mathbb{P}^n$:

$$\bar{\Phi}[a_1 : \dots : a_n] = \left[-\frac{1}{\beta_0} \sum_{i>0} \beta_i a_i : a_1 : \dots : a_n \right].$$

Its image is equal to H , and it is injective by considering the last n coordinates in the image of $\bar{\Phi}$. So it restricts to a bijective morphism $\bar{\Phi}: \mathbb{P}^{n-1} \rightarrow H$.

Now, Φ' 's image is $Z_{\mathbb{A}}(h) - 0$ and it is injective too. Its inverse is given by $(\Phi')^{-1}(a_0, \dots, a_n) = (a_1, \dots, a_n)$ which is a morphism of varieties as each component is regular. Taking inverses, we have the commutative diagram

$$\begin{array}{ccc} \mathbb{A}^{n-1} - 0 & \xrightarrow{(\Phi')^{-1}} & Z_{\mathbb{A}}(h) - 0 \\ \pi \downarrow & & \downarrow \pi \\ \mathbb{P}^n & \xrightarrow{\bar{\Phi}^{-1}} & Z_{\mathbb{P}}(h) \end{array}$$

and since the π s are quotients, this means $\bar{\Phi}^{-1}$ is a morphism as well. Thus $\bar{\Phi}$ is an isomorphism as desired.

- (2) The line connecting $[b_0 : \dots : b_n]$ and $[a_0 : \dots : a_n]$ is given by the points of the form $[sb_i + ta_i]_i$. This is in H iff

$$0 = h[sb_i + ta_i]_i = h(sb_i + ta_i)_i = sh(b) + th(a),$$

so we have that $t = -sh(b)/h(a)$ so the resulting point is

$$\left[sb_i - s \frac{h(b)}{h(a)} \right]_i = \left[b_i - \frac{h(b)}{h(a)} \right]_i,$$

as desired.

The coordinates of φ are all polynomial in b , so it is a morphism of varieties.